

Post-Quantum Cryptography (PQC) Migration & Risk Report

Comprehensive Multi-Engine Assessment: ML Risk Classifier, HNDL Threat, Mosca Timeline & Mitigation Roadmap

ASSESSMENT TARGET

ECDAT Enterprise Post-Quantum Cryptographic Risk Report

GENERATED AT

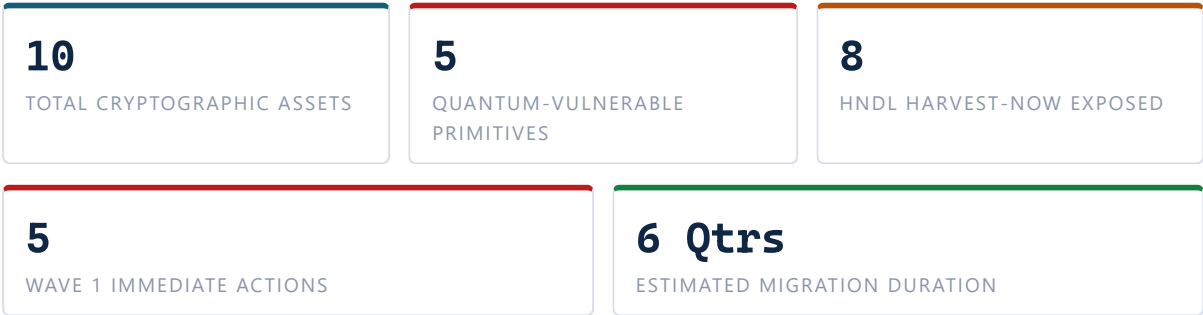
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PQC STANDARDS

NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA)

CONFIDENTIAL · ENTERPRISE SECURITY GOVERNANCE REPORT

1 Executive KPI Dashboard



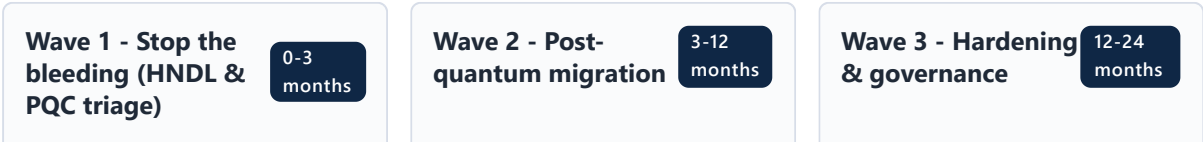
Executive Summary: The ECDAT Enterprise Post-Quantum Cryptographic Risk Report evaluated a total of 10 cryptographic assets, identifying 6 high-risk configurations and 5 quantum-vulnerable items concentrated within a single internal blast zone. Although zero urgent or critical risk issues were detected immediately, proactive remediation is vital over an estimated 6-quarter timeline. All vulnerable primitives must undergo systematic migration to NIST-approved post-quantum standards according to structured waves.

2 Quantum Threat & Timeline Deficit

Under **Michele Mosca's Theorem** ($X + Y > Z$), if the migration duration (X) plus the required data shelf-life (Y) exceeds the estimated time to a Cryptographically Relevant Quantum Computer (Z), assets fall into an active security deficit.

Quantum Vulnerability Vector: Public-key primitives such as RSA and ECC are fundamentally vulnerable to Shor's algorithm running on a future cryptographically-relevant quantum computer (CRQC). Furthermore, ciphertext captured today remains susceptible to Harvest-Now-Decrypt-Later (HNDL) attacks, making timely post-quantum migration an urgent enterprise imperative.

3 Wave-Based PQC Remediation Roadmap



Rotate HNDL-exposed keys, add hybrid handshakes and stage urgent post-quantum replacements for publicly exposed primitives.

Assets Allocated: 5

Systematically move public-key signatures and key exchange to ML-KEM / ML-DSA per CNSA 2.0, covering downstream trust chains.

Assets Allocated: 1

Consolidate symmetric modes (AES-256), strengthen hashes to SHA-3, retire legacy ciphers and codify crypto policy so new failures cannot ship.

Assets Allocated: 4

4 Inventory & Target PQC Replacements

ASSET ID	ALGORITHM	CATEGORY	MIGRATION WAVE	TARGET PQC STANDARD	BLAST RADIUS	ACTION DIRECTIVE
DISC-001	RSA-2048	PUBLIC_KEY	Wave 1	ML-KEM-768 / ML-DSA-65	CRITICAL	Replace RSA-2048 with ML-KEM-768 / ML-DSA-65 (CNSA 2.0 / NIST PQC) in 'unknown'.
DISC-002	ECDSA	PUBLIC_KEY	Wave 1	ML-DSA-65	CRITICAL	Replace ECDSA with ML-DSA-65 (CNSA 2.0 / NIST PQC) in 'unknown'.
DISC-006	RSA-3072	PUBLIC_KEY	Wave 1	ML-KEM-768 / ML-DSA-65	CRITICAL	Replace RSA-3072 with ML-KEM-768 / ML-DSA-65 (CNSA 2.0 / NIST PQC) in 'unknown'.
DISC-007	RSA-4096	PUBLIC_KEY	Wave 1	ML-KEM-1024 / ML-DSA-87	CRITICAL	Replace RSA-4096 with ML-KEM-1024 / ML-DSA-87 (CNSA 2.0 / NIST PQC) in 'unknown'.
DISC-008	ED25519	PUBLIC_KEY	Wave 1	ML-DSA-65 / SLH-DSA	CRITICAL	Replace ED25519 with ML-DSA-65 / SLH-DSA (CNSA 2.0 / NIST PQC) in 'unknown'.
DISC-004	3DES-CBC	SYMMETRIC	Wave 2	AES-256 (GCM)	HIGH	Maintain 3DES-CBC with AES-256 strength; re-validate the cipher mode and key-management rotation in '{service}'.
DISC-009	AES-128-GCM	SYMMETRIC	Wave 3	AES-256 (GCM)	MEDIUM	Maintain AES-128-GCM with AES-256 strength; re-validate the cipher mode and key-management rotation in '{service}'.
DISC-003	AES-256-GCM	SYMMETRIC	Wave 3	AES-256 (GCM)	LOW	Maintain AES-256-GCM with AES-256 strength; re-validate the cipher mode and key-management rotation in '{service}'.
DISC-010	X25519+ML-KEM-768	UNKNOWN	Wave 3	Review & standards triage	LOW	Triage the X25519+ML-KEM-768 reference in 'unknown' for standards compliance.
DISC-005	ML-KEM-768	UNKNOWN	Wave 3	Review & standards triage	MEDIUM	Triage the ML-KEM-768 reference in 'unknown' for standards compliance.

5 Strategic Governance & Policy Recommendations

Wave 1 - Stop the bleeding (HNDL & PQC triage)

Inventory and prioritize all quantum-vulnerable assets within the internal 'unknown' service blast zone, establishing baseline visibility and halting new deployments of legacy algorithms.

Wave 2 - Post-quantum migration

Execute systematic cryptographic replacements for identified vulnerable public-key and symmetric assets, transitioning to NIST FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA) standards.

Wave 3 - Hardening & governance

Enforce strict cryptographic governance, continuously audit algorithm usage via automated tooling, and complete long-term hardening across all enterprise services.